26. Volume & 3D Shapes

Year 4 · **Supplementary Worksheet** — extra practice & worked breakdowns

Worked Examples

Example 1 — Counting unit cubes:

A cuboid is built from centimetre cubes. It is 4 cubes long, 2 cubes wide and 3 cubes high. How many cubes make it up?

Answer: 4 × 2 × 3 = **24 cubes**.

Example 2 — 3D shape facts:

How many faces, edges and vertices does a cube have?

Answer: A cube has **6 faces, 12 edges and 8 vertices**.

Example 3 — Volume formula:

What is the volume of a cuboid measuring 5cm × 4cm × 2cm?

Answer: Volume = length × width × height = 5 × 4 × 2 = **40 cm³**.

Example 4 — Finding a cube's side from its volume:

A cube has a volume of 27 cm³. What is the length of one side?

Answer: 27 = 3 × 3 × 3, so the side length is **3 cm**.

Example 5 — Multi-step word problem:

A box measuring 4cm × 2cm × 3cm is completely filled with 1cm cubes, then a third of them are removed. How many cubes are left?

Answer: The box holds 4 × 2 × 3 = 24 cubes. A third of 24 is 8, so 24 − 8 = **16 cubes** are left.

Foundation

15 questions

1. What is the volume of a cuboid measuring 4cm × 2cm × 2cm?

2. How many faces does a cube have?

3. Which 3D shape has 6 faces, 12 edges and 8 vertices? A) sphere B) cube C) cone D) cylinder

4. How many edges does a cube have?

5. How many vertices does a cube have?

6. What is the volume of a cuboid measuring 5cm × 2cm × 1cm?

7. Which shape has NO flat faces at all? A) cube B) cuboid C) sphere D) cone

8. What is the volume of a cube with sides of 3cm?

9. Select ALL of these 3D shapes that have at least one curved surface: A) sphere B) cube C) cylinder D) cuboid E) cone

10. How many faces does a cuboid have?

11. What is the volume of a cuboid measuring 4cm × 1cm × 3cm?

12. Which shape is a cube an example of? A) a sphere B) a type of cuboid C) a cylinder D) a cone

13. How many edges does a cuboid have?

14. What is the volume of a cuboid measuring 5cm × 1cm × 1cm?

15. How many vertices does a cuboid have?

Standard

20 questions

1. A cuboid measures 5cm × 3cm × 2cm. What is its volume?

2. Which cuboid has a volume of 30cm^3? A) 5×3×2 B) 6×2×2 C) both A and B D) neither

3. A cuboid measures 4cm × 2cm × 3cm. What is its volume?

4. How many faces does a triangular prism have?

5. How many vertices does a triangular prism have?

6. Which 3D shape has exactly 2 flat circular faces and 1 curved face? A) sphere B) cylinder C) cone D) cube

7. A cube has a volume of 27cm^3. What is the length of one side?

8. Select ALL of these cuboids that have a volume of MORE than 15cm^3: A) 2×2×2 B) 3×2×2 C) 4×2×2 D) 5×1×2 E) 3×3×2

9. How many edges does a square-based pyramid have?

10. How many vertices does a square-based pyramid have?

11. A cuboid measures 5cm × 2cm × 4cm. What is its volume?

12. Which shape has 5 faces, 8 edges and 5 vertices? A) triangular prism B) square-based pyramid C) cube D) cuboid

13. A cube has sides of 4cm. What is its volume?

14. How many faces does a cone have (flat and curved together)?

15. A cuboid measures 6cm × 3cm × 2cm. What is its volume?

16. A cube has a volume of 64cm^3. What is the length of one side?

17. A cuboid measures 7cm × 1cm × 2cm. What is its volume?

18. Which cuboid has a volume of 16cm^3? A) 4×2×2 B) 8×1×2 C) both A and B D) neither

19. Select ALL of these cuboids that have a volume of MORE than 15cm^3: A) 3×2×3 B) 2×2×4 C) 5×1×2 D) 4×1×4 E) 2×3×2

20. A cuboid measures 9cm × 5cm × 1cm. What is its volume?

Hard

5 questions

1. A cuboid has a volume of 36cm^3, a length of 6cm and a width of 2cm. What is its height?

2. A box measuring 4cm × 3cm × 2cm is completely packed with 1cm cubes. Select ALL of the statements below that are TRUE: A) the box holds 24 small cubes B) the box's volume is 24cm^3 C) the box holds 18 small cubes D) each small cube has a volume of 1cm^3 E) the box's volume is 18cm^3

3. Comparing two cuboids — Cuboid A at 5×2×4cm and Cuboid B at 4×5×2cm — which one takes up more volume?

4. A cuboid has a volume of 45cm^3, a length of 5cm and a height of 3cm. What is its width?

5. A cube has a volume of exactly 27cm^3. What is the length of one edge? A) 2cm B) 3cm C) 6cm D) 9cm

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